Weiqi Gai

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Research Interests

Autonomous robot navigation and embodied intelligence: motion planning, semantic exploration, and spatial reasoning in unknown environments; vision-language-action (VLA) and world action models (WAM) for generalizable decision-making.

Education & Research Appointments

Differential Robotics (Hangzhou) Co., Ltd. Aug. 2025 – Present
Large-Model Algorithm Intern — Intelligent UAV Navigation

Beihang University Sept. 2024 – Present
Master's Student — Aircraft Design — Intelligent UAVs Advisor: Prof. Zhijun Meng

Zhejiang University Huzhou Institute, FAST-LAB June 2024 – Sept. 2024
Research Assistant — Autonomous UAV Exploration Advisor: Prof. Fei Gao

Beihang University Sept. 2020 – June 2024
Undergraduate — Aircraft Environment Control and Life Support

Publications & Manuscripts

1. **OSET: Fully Onboard Semantic Exploration for UAVs with Traversability-Aware Scene Graph**
   Gai, W., Gao, Y., Zhou, Y., Wu, Y., Zhou, X., Zhao, G., Wang, J., Gao, F. & Meng, Z.
   Chinese Journal of Aeronautics (CJA), 2026, ***Under Review****.*

   Traversability-aware scene graphs, onboard LLM reasoning, and frontier planning enable semantic UAV exploration entirely on an Orin NX. Achieved relative gains of **54% in success rate** and **65% in SPL** on long-horizon Unity tasks.

2. **NavDreamer: Video Models as Zero-Shot 3D Navigators**
   Huang, X.†, **Gai, W.**†, Wu, T., Wang, C., Zheng, Q., et al.
   IEEE Robotics and Automation Letters (RA-L), 2026, ***Accepted****.* †*Co-first authors.*

   VLM-ranked video predictions, inverse dynamics, and metric depth yield zero-shot 3D navigation waypoints. Achieved **∼85% success** across **20 real-world indoor/outdoor task categories**; reduced outdoor relative-scale error from **54% to 6%**.

3. **Grounding Human Interventions via Anchored Corrective-Measure Estimation**
   Cui, X., Zhu, M., **Gai, W.**, Wu, Y., Zhou, X., Gao, F., Wang, B., Zhou, W. & Li, H.
   AAAI, 2027, ***Under Review****.*

   Risk-aware corrective weighting and context-aware optimal transport reuse sparse human interventions. Evaluated on three simulation benchmarks and real ground robots/quadrotors: **+34.3 percentage points** in average success over PPL, with **25.3% less expert supervision**.

4. **SwiftGuide: Observation-Preserving Hierarchical Multimodal Semantic Mapping for User-Aligned Embodied Navigation Agents**
   Huang, Z., **Gai, W.**, Zhou, X., Wu, Y., Gao, F. & Lyu, X.
   Manuscript, 2026, ***In Preparation****.*

   Hierarchical visual memory, personalized VLM retrieval, and HiGVD planning enable object retrieval and return navigation. Across **81 explorations/965 queries**, description Hit@5 reached **60.8%** (best baseline: 39.0%); navigation success was **84.6%** on 267 reachable targets (+8.6 percentage points over the best baseline).

5. **Efficient Autonomous UAV Exploration Framework With Limited FOV Sensors for IoT Applications**
   Wang, Z., Meng, Z., Tian, T., **Gai, W.**, Zhao, G., et al.
   IEEE Internet of Things Journal, 12(1): 713–725, 2025, ***Accepted****.*

   Frontier generation, anisotropic viewpoint sampling, sliding-window TSP, and active-perception trajectory optimization reduced exploration time by **17–27%** and path length by **10–17%** in simulation and real-world experiments.

Research & Engineering Projects

OSET: Fully Onboard Semantic Exploration for UAVs Aug. 2025 – May 2026
Differential Robotics / Project Lead; First Author

- Independently led problem formulation, system design, core implementation, UAV integration, real-flight experiments, and manuscript writing for fully onboard semantic exploration (Publication 1).
- Transferred core modules into multiple company projects and operational product demonstrations.

Navigation World Model: Inference Acceleration & Deployment July – Sept. 2026
Differential Robotics / Core Developer

- Built an **SFT → SCM/RCM warmup → DMD** pipeline for navigation-video generation; reduced denoising from **30 to ∼4 steps** using distillation, sparsification, and a lightweight TAE.
- Reduced generation time from **171.568 s to 0.702 s (244.4×)** while preserving navigation semantics and temporal continuity, supporting low-latency deployment.

Qwen3.5 Post-Training & Evaluation (Mathematics) Sept. 2026
Research and Learning Project

- Compared Attention LoRA, Attention+MLP LoRA, and full SFT using **Megatron Bridge/Core** on **4 NVIDIA H20 GPUs**; curated and deduplicated **500,000** training examples.
- Built parallel **vLLM/HTTP** evaluation on **1,819 GSM8K/Math500 questions**; best-checkpoint accuracy rose from **41.4% to 55.1%** (Direct) and **36.3% to 60.0%** (CoT).

LeRobot: Real-Robot Manipulation Learning 2026
Research and Learning Project / Real-Robot Reproduction

- Collected **200 teleoperated demonstrations** for XT60 connector grasping; reproduced **ACT, Diffusion Policy, π_0, and $\pi_{0.5}$**, covering preprocessing, training, deployment, and closed-loop visual grasping.

FlyAgent: General-Purpose Embodied Flight Agent Nov. 2025 – Present
Differential Robotics / Core Developer

- Designed system architecture, communication topology, and onboard infrastructure connecting web dialogue, agent services, task orchestration, and flight control.
- Extended **CoPaw** and debugged tool calling for navigation, manipulation, exploration, tracking, and inspection; achieved stable operation, with productization underway.

Large-Scale Heterogeneous Robot Swarm Exploration June – Sept. 2024
Zhejiang University Huzhou Institute, FAST-LAB / Core Developer

- Developed a **15-robot swarm** spanning vision/LiDAR quadrotors, coaxial UAVs, and radar-equipped ground vehicles; led UAV exploration, inter-robot map exchange, and hardware debugging.
- Adapted **FUEL** with local grids, a global auxiliary grid, and topological paths to reduce dependence on dense global maps in multistory, obstacle-rich environments.

Autonomous Obstacle Avoidance for an Unmanned Helicopter Dec. 2024 – June 2025
AVIC Institute 602 / Project Lead

- Led requirements, scheduling, algorithm adaptation, and platform-specific deployment; customized a multirotor demonstrator for integrated perception–planning–control validation. Delivered on schedule.

Honors & Awards

- Outstanding Graduate Student, Beihang University (2026).
- Master's Scholarships, Beihang University: Second Class (2026); First Class (2025).
- Meritorious Winner (M Award), Mathematical Contest in Modeling / Interdisciplinary Contest in Modeling.

Technical Skills & Languages

Software: C++, Python, ROS, MATLAB; Megatron, vLLM; AI Harness workflows.
Robotics & AI: Motion planning, autonomous exploration, LLM/VLM post-training, world-model distillation.
Hardware: SolidWorks, Altium Designer; UAV structural/PCB design, flight-controller tuning, system integration.
English: IELTS 6.5; CET-4/6 passed.